

PROJECT EXECUTION BLUEPRINT

AI-Driven Operational Management System

OBJECTIVE

To design and build a centralized operational system that enables:

- Real-time project visibility
- Structured task execution
- Resource optimization
- Financial awareness
- Future-ready AI-driven decision support

The system will evolve in 4 structured phases, ensuring stability before intelligence.

PHASE 1: OPERATIONAL VISIBILITY (FOUNDATION)

Goal:

Establish a reliable execution system where projects, employees, and tasks are managed in a structured and traceable manner.

This phase forms the backbone of the entire platform.

1. ADMIN PANEL

A centralized control system for managing operations.

A. Executive Dashboard

Purpose:

Provide leadership with real-time operational insights.

Core Metrics:

- Projects at risk
- Delayed tasks
- Developer utilization
- Completion trends
- Budget burn readiness

B. Project Management System

Capabilities:

- Create and manage projects
- Dedicated project detail dashboard

Project Detail Includes:

- Project timeline
- Employee assignment
- Expense tracking
- Task ledger (all tasks under project)
- SRS document upload (storage only in this phase)

Metadata Stored:

- Project owner
- Start & end date
- Priority
- Budget placeholder
- Status

Lifecycle States:

- Planned
- Active
- Blocked
- Completed

Additional Capabilities:

- Advanced search & filtering
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C. Employee Management System

Capabilities:

- Create and manage employees
- Enable / disable employee accounts
- Admin-side data control

Employee Profile Includes:

- Role assignment (RBAC ready)
- Skill tagging
- Availability tracking
- Cost rate (used for financial intelligence later)

Note:

Designed to support operational allocation, not full HRMS complexity.

D. Financial Ledger (Operational Level)

Purpose:

Provide a unified view of cost-related activity.

Capabilities:

- Aggregated task-level financial tracking
- Filterable by:
 - Project
 - Employee
 - Time period

Note:

This is not a full accounting system. It is an operational cost tracker.

2. EMPLOYEE PLATFORM

A working interface for employees to interact with the system.

Capabilities:

A. Employee Dashboard

- Assigned tasks overview
- Status tracking
- Activity visibility

B. Task Submission System

- Coding task form
- Digital marketing task form
- Structured input for task updates

C. Profile Section

- Basic personal details
- Operational data (role, assignment context)
- Document upload (verification support)

D. Task History

- Complete record of past work
- Status and activity tracking

PHASE 1 OUTCOME

At the end of this phase, the system will:

- Act as a central operational hub
- Track all active work across projects
- Provide leadership-level visibility
- Maintain structured execution data

This phase ensures strong data integrity for future intelligence layers.

PHASE 2: INTEGRATIONS LAYER

Goal:

Introduce measurable execution tracking and operational signals without adding heavy system complexity.

1. TIME LOGGING SYSTEM

Capabilities:

- Manual time entry
- Attach time logs to tasks/assignments
- Edit logs with history tracking
- Optional approval workflow

Outcome:

Creates the foundation for productivity and cost analysis.

2. BUDGET CONSUMPTION INDICATOR

Purpose:

Provide a clear view of budget utilization.

Logic:

- Based on time logged × employee cost rate

Output:

- Budget vs consumed budget
- Visual indicators for project health

3. RULE-BASED ALERT SYSTEM

Purpose:

Enable proactive monitoring of risks.

Initial Rules:

- Task delay beyond threshold → alert manager
- Developer utilization > 90% → warning
- Project nearing deadline with low completion → risk flag

Note:

Rules are system-defined initially and can be expanded later.

4. CLIENT MANAGEMENT

Capabilities:

- Maintain client profiles
- Track associated projects
- View project status per client
- Basic profitability visibility

5. PROJECT COST TRACKING (DETAILED VIEW)

Capabilities:

- Track all cost components within a project
- Aggregate cost visibility
- Link costs with tasks and resources

6. ATTENDANCE SYSTEM (LIGHTWEIGHT)

Capabilities:

- Employee attendance tracking
- Admin visibility

Note:

Kept minimal to avoid HR system complexity.

PHASE 2 OUTCOME

- Measurable execution tracking
- Cost visibility at project level
- Early risk detection system
- Improved operational control

PHASE 3: INTELLIGENCE LAYER

Goal:

Transform structured data into actionable intelligence using AI.

1. SRS INGESTION PIPELINE

Flow:

Upload → Parse → Process → Store

Capabilities:

- Document ingestion
- Content structuring
- Preparation for AI processing

2. AI-ASSISTED TASK EXTRACTION

Capabilities:

- Convert SRS into structured tasks
- Output in reviewable format
- Human approval before task creation

3. EMBEDDINGS & RETRIEVAL ARCHITECTURE

Capabilities:

- Vector representation of documents
- Semantic search
- Context retrieval for AI features

4. PROGRESS ESTIMATION SYSTEM

Capabilities:

- Estimate task/project completion
- Based on:
 - Historical data
 - Time logs
 - Task patterns

5. COMPLETION INDICATORS

AI evaluates:

- Task updates
- Time logs
- Activity signals
- Dependencies

Output:

- Completion confidence
- Risk indicators

6. PROMPT ENGINEERING & EVALUATION LOOP

Capabilities:

- Controlled AI outputs
- Continuous improvement of results
- Accuracy monitoring

PHASE 3 OUTCOME

- Intelligent task understanding
- Predictive progress tracking
- AI-supported operational insights

PHASE 4: AI PROJECT MANAGER

Goal:

Evolve the system into a proactive decision-support engine.

1. DUPLICATE WORK DETECTION

Capabilities:

- Identify similar or redundant tasks
- Prevent duplicated effort

2. WORKLOAD INSIGHTS

Capabilities:

- Identify over-utilized resources
- Detect uneven distribution
- Highlight idle capacity

3. TASK RECOMMENDATION ENGINE

AI Suggestions:

- Task reassignment
- Deadline adjustments
- Task splitting
- Priority optimization

4. RISK DETECTION SYSTEM

Capabilities:

- Detect delays and bottlenecks
- Identify high-risk projects
- Highlight execution gaps

5. BEHAVIORAL PATTERN ANALYSIS

Capabilities:

- Analyze team working patterns
- Identify inefficiencies
- Improve planning insights

6. AI FEEDBACK LOOP

Capabilities:

- Continuous learning from system data
- Improved recommendations over time

PHASE 4 OUTCOME

- Proactive decision support
- Reduced operational risk
- AI-assisted project management
- Strategic execution optimization

FINAL NOTE

The system is designed with a layered approach:

Phase 1 → Build structure

Phase 2 → Add measurement

Phase 3 → Introduce intelligence

Phase 4 → Enable autonomous insights

This ensures scalability, stability, and long-term value.